

Why PFAS half-lives differ between humans, rats and mice

A printable summary of the evidence, the assumptions behind the published numbers, and what is still unresolved

Scope	186 full texts read; 24 per-paper extractions; 1,438 database rows
Full review	report/REPORT.md, ~2,002 lines, 34 runnable scripts, 16 figures
Repository	github.com/sdey17/pfas_tk
Provenance	every database row carries a PMID or DOI and the table it was read from; values computed in this work are marked as such

The question

Published PFAS serum half-lives disagree by orders of magnitude - between humans and animals, between rat and mouse, between male and female, and between papers describing the same experiment. This work asks why, and answers it by re-deriving the numbers from the tables they came from rather than from the sentences written about them.

One identity governs the whole subject:

$$t_{1/2} = \ln 2 \cdot V_d / CL$$

Half-life is not a property of a chemical. It is a ratio of how much of the body the chemical occupies to how fast the body removes it, and any two of the three terms fix the third. Nearly every disagreement in this literature turns out to be about **which two were measured and which one was assumed** - and in the most consequential case, the assumed one is the number everybody quotes.

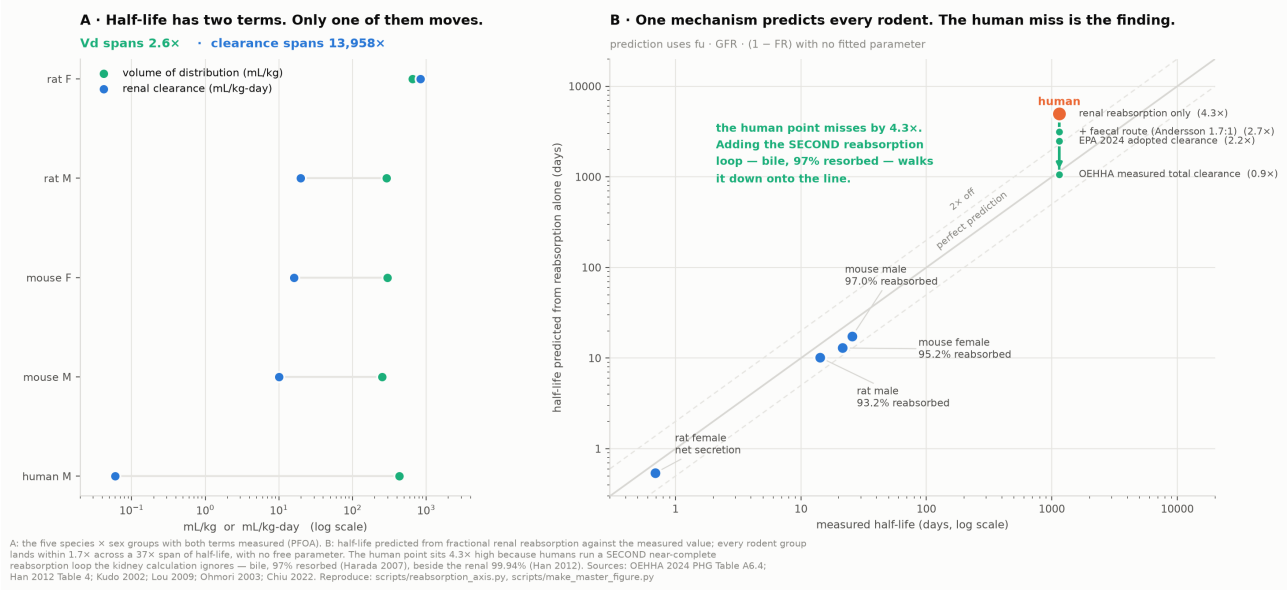
What is in here

- 1 The species difference is a clearance difference, and it is concentrated in females
- 2 One mechanistic axis orders every species: fractional renal reabsorption
- 3 Exposure level is related to neither half-life nor volume of distribution
- 4 The human volume of distribution that regulation rests on is an assumption
- 5 Five sixths of the field is empty
- 6 Humans run two near-complete reabsorption loops, not one
- 7 Half-life is the wrong endpoint for a structure-activity model

then: what this corrects in the published record, what is still unresolved, and how to check any of it

The whole argument in one figure

PFAS half-lives differ between species because of how much is reabsorbed, not how much is distributed



Left: across five species-by-sex groups, the volume of distribution spans 2.6× while renal clearance spans 13,958×. The variance is in clearance, not distribution. **Right:** half-life predicted from fractional renal reabsorption alone, against observation, on log axes with the 1:1 line. Every rodent point falls within 1.7× over a 37× span, with no fitted parameter. The human point sits 4.3× high, and the walk-down beneath it - renal only 4.3×, plus faecal 2.7×, EPA clearance 2.2×, OEHA measured clearance 0.9× - is what led to the second reabsorption loop in finding 6. Regenerate with `scripts/make_master_figure.py`.

The seven findings

1. The species difference is a clearance difference, and it is concentrated in females

Six datasets measure *both* terms in *both* sexes inside single experiments - three species, two compounds, four laboratories, no cross-study comparison required.

	V _d male/female	clearance female/male
span across all six datasets	1.33 - 2.18×	0.78 - 44.3×
	a 1.6× spread	a 56× spread

The distribution ratio is male-higher every single time and never leaves a narrow band. The clearance ratio ranges over nearly two orders of magnitude **and changes sign between species**. Argoul 2026 reaches the same conclusion inside one experiment: across 11 PFAS dosed as a single cocktail, clearance spans 5,254× while V_{ss} spans 7.7×. Taking both limbs from primary sources, female rat divided by female mouse PFOA clearance is 373-496×, against 7.0× in males. Tatum-Gibbs 2011 replicates the structure in a second compound with strains matched.

2. One mechanistic axis orders every species: fractional renal reabsorption

$$CL_{renal} = f_u \cdot GFR \cdot (1 - FR)$$

species	sex	GFR (L/d/kg)	renal clearance (mL/d/kg)	reabsorbed
human	both	2.57	0.03	99.94%
mouse	female	16.7	16	95.2%
mouse	male	16.7	10	97.0%
rat	female	14.4	666	net secretion
rat	male	14.4	18.2	93.7%
Japanese macaque	female	8.5	32	81.2%
Japanese macaque	male	8.5	15	91.2%
dog	female	5.3	50.8	52%
dog	male	5.3	43	59%
rabbit	female	4	670	net secretion
rabbit	male	4	640	net secretion

Read from Han 2012 Table 4 at source (db/primary_2026/han2012_table4_reabsorption_axis.csv). Two values differ from the widely-cited adaptation of this table - see the corrections overleaf.

The axis reproduces rodent half-lives within $1.7\times$ over a $37\times$ span with no free parameters, and an unrelated dataset recomputes mouse PFOA at 95.9%, inside the 95.2-97.0% already in use. **But it has two factors, not one.** Humans have the longest half-life yet reabsorb *less* in absolute terms (51 mL/d/kg) than male rats (270) or mice (318-324). Of the $333\times$ male-mouse-to-human renal clearance gap, the escape fraction carries $50\times$ and the six-fold lower human GFR carries $6.5\times$ - 68% and 32% on a log scale. Reabsorption is the larger term, which is why the single-axis framing works; but a third of the difference is filtration rate, which no transporter story explains.

The mechanism does not transfer, even though the fraction does. A candidate transporter must be both sex-divergent and able to carry PFOA. In the rat only Oatp1a1 is both ($23\times$ male-predominant, androgen-induced, transports C8-C10); Oat2 is strongly sex-divergent but carries no PFOA, and Oat1/Oat3 carry it but are not sex-divergent. In humans, OATP1A2 - the closest orthologue of rat Oatp1a1 - does not transport PFOA at all; human apical reabsorption runs through OAT4 and URAT1, neither androgen-regulated.

3. Exposure level is related to neither half-life nor volume of distribution

Volume-of-distribution slopes against dose run -0.21 to +0.25 with inconsistent sign. Four independent dose slopes for half-life cluster at +0.08 to +0.12 against a mechanistic ceiling of 1. The between-person association does not survive age adjustment: in Li 2022's own variance decomposition, age carries $2\text{-}13\times$ the partial R^2 of initial PFAS concentration, and the exposure-tertile effect is 25-37% the size of the age effect. One real exception: the female rat, slope +0.21 over a $3,200\times$ dose range, consistent with saturable *secretion*; the male rat is -0.02.

4. The human volume of distribution that regulation rests on is an assumption, not a measurement

Nine of 42 adopted regulatory clearance factors are computed from Thompson 2010's 170 mL/kg. That paper's supplementary table is headed "*Input data for the calibration of the Vd parameter*", with a final column headed "*calculated Vd*" - and its published values reproduce to three figures from intake, serum and an assumed elimination rate taken from a study in **the same two communities**.

The consequence is sharper than "circular". Forming $CL = \ln 2 \cdot V_d / t_{1/2}$ makes the half-life **cancel exactly**, leaving $CL = \text{Dose/Serum} = 0.132\text{-}0.138$ mL/kg-day. But V_d does not cancel: it is proportional to whatever half-life is assumed.

half-life assumed	V_d that follows	derived clearance
2.3 y (the value used)	168 mL/kg	0.132-0.138 mL/kg-day
3.8 y	277 mL/kg	0.132-0.138 mL/kg-day (unchanged)

So adopting "170" alongside a different half-life silently contradicts the data it came from. Reproduced from the paper's own Table S1 to within $1.000\text{-}1.003\times$ by scripts/thompson_vd_circularity.py. Meanwhile every direct measurement (74, 121, 113-199 mL/kg) falls *below* the assigned range, and the one population fit (430) sits far above it.

5. Five sixths of the field is empty

39 chemicals × 11 species × 4 parameters = 1,716 cells. **1,466 have no data at all (85%)**; a further 94 rest on a single study. Even PFOS, the best-covered compound in the world, fills 29 of 44. By species: human 71, rat 61, mouse 45, monkey 28 - then **dog 2**. Mining four further compilations moved coverage from 143 to 156 cells, so the hole is real rather than a search artefact.

6. Humans run two near-complete reabsorption loops, not one

The renal axis leaves a residual: predicted human half-life comes out 4.3× short of observation. The missing term is enterohepatic. Harada 2007 sampled serum and bile from four gallstone-surgery patients and measured a biliary resorption rate of 0.97. Bile PFOS (27.9 ng/mL) actually *exceeds* serum (23.2), so bile is a real excretion route - but 97% of what is secreted is reabsorbed from the gut.

loop	fraction reabsorbed	source
renal (PFOA)	99.94%	Han 2012 Table 4
biliary (PFOS)	97%	Harada 2007, n = 4

Both are near-unity, both lengthen the human half-life, and **both are interruptible** - which is why bile-acid sequestrants work. Delaere 2025: treated PFOS half-life 1.2 y (n = 19) against 7.3 y under observation (n = 9), a 6.1× difference; PFHxS 2.5 y against 9.4 y. Genuis 2010 and Møller 2024 agree in direction.

It also reconciles the faecal-route dispute without either side being wrong. At 97% resorption, **gross** biliary flux is large while **net** faecal elimination is small - so Andersson, measuring faeces under ongoing intake, saw a large signal, and Abraham, following a labelled bolus, saw almost nothing leave that way. Different quantities, one reconciling constant. The caveat is that n = 4, and the number is now load-bearing.

7. Half-life is the wrong endpoint for a structure-activity model

Because $t_{1/2} = \ln 2 \cdot V_d / CL$, half-life carries a glomerular filtration term that differs 6.5× between mouse and human with no change in chemistry. A structural model fitted to it is being asked to absorb body size. The replacement is the dimensionless renal handling ratio:

$$R = CL_{renal} / (f_u \cdot GFR)$$

$R < 1$ is net reabsorption, $R > 1$ net secretion, and $R = 1 - FR$, so it is finding 2's axis on a scale that neither saturates near 1 nor runs unboundedly negative. On the nine compounds where both terms were measured in one experiment (Argoul 2026, male mouse), R spans 875×:

compound	F-carbons	head group	ether O	f _u %	log ₁₀ R	handling
PFDA	9	-COOH	0	0.42	-2.509	reabsorbed
PFHxS	6	-SO ₃ H	0	1.3	-2.344	reabsorbed
PFNA	8	-COOH	0	0.35	-1.766	reabsorbed
PFOA	7	-COOH	0	0.87	-1.384	reabsorbed
PFOS	8	-SO ₃ H	0	0.25	-1.305	reabsorbed
PFBA	3	-COOH	0	77.0	-0.993	reabsorbed
GenX	5	-COOH, ether	1	26.0	-0.985	reabsorbed
PFO2OA	5	-COOH, ether	2	10.0	0.186	SECRETED
PFHxA	5	-COOH	0	25.0	0.433	SECRETED

Built by scripts/qsar_endpoint_table.py into db/qsar/qsar_endpoint_table.csv. "F-carbons" counts carbons bearing fluorine, which excludes a carboxylate's acid carbon and includes every carbon of a sulfonate.

Three results constrain any QSAR. **Chain length alone does not order the endpoint** - within the carboxylates the series is non-monotonic and PFHxA crosses into net secretion, so a model using carbon number as its only descriptor is already falsified on nine compounds. **The head group carries about 3× at matched chain length** (PFNA against PFOS, both 8 fluorinated carbons, 2.9× apart). **Ether oxygens move the endpoint 1.4 log units at constant chain length, non-monotonically**: PFHxA with none is secreted, GenX with one is reabsorbed, PFO2OA with two is secreted - so the replacement chemicals sit on both sides of the divide.

The same construction sharpens finding 4's successor problem. R is linear in $1/f_u$, so the three human free fractions in the literature give three different answers for human PFOA:

f_u	source	$\log_{10} R$	gap to male mouse
0.10	PBPK models reading ">90% bound" as "about 90%"	-3.93	354×
0.02	Han 2012's stated assumption	-3.23	71×
0.00061	Fischer, measured at physiological ligand:protein	-1.72	2.2×

Under the measured value the species difference is almost entirely **binding**, which would mean the transporter literature is explaining a quantity that barely differs between the species. This is stated as a sensitivity rather than a result, because the mouse and human free fractions come from different methods at different ligand:protein ratios - the exact artefact diagnosed below. One panel measuring f_u for these compounds in mouse, rat and human plasma by a single method would settle it, and needs no animals.

What this corrects in the published record

Each entry is traced to the table or figure that contradicts it. The full list of 18 is section 8 of the report.

source	correction
EPA	the rat Oatp1a1 male/female ratio quoted as "2.5-fold" is a <i>different transporter's</i> number (OAT-K); the primary value is 23×
OEHHA	its adaptation of the source reabsorption table altered two values (human 99.94 to 99.8, male rat 93.7 to 93.2)
Provenance	the $f_u = 0.02$ assumption behind the axis is Han 2012's, not OEHHA's, and Han calls its own values "rough estimates"
Andersson 2025	transposes Thompson's PFOA and PFOS volumes (it is PFOA 170, PFOS 230); an earlier claim in this work that Zhang 2013 did so was itself wrong and is withdrawn
Cheng 2006	its abstract calls renal Oatp1a1 "female-predominant", then reports androgens <i>increase</i> it and concludes androgens are the exclusive cause - self-contradictory, and against Cheng 2005
Metabolic inertness	"PFAS are metabolically inert" fails for 6:2 Cl-PFESA, which is biotransformed (Yi 2022)
This work	its own refits of the EPA raw curves compress sex ratios about 13× - see below

The self-correction matters most

The terminal-slope fits in db/cpheafitted_halfives.csv select the window with the best adjusted R^2 , which on a biphasic curve is the shallow tail. Against a published value for the same experiment the fit returns **1.44 d against 0.08 d** for the female rat, and compresses a **71× sex ratio to 5.6×**. The bias runs one way, so every conclusion above survives and several strengthen - but that column must not be read as comparable to published half-lives. It carries a tail_selection_flag and is kept rather than deleted, because it remains valid for within-curve comparison.

What is still unresolved

1. A measured human transport-affinity constant

Whether reabsorptive transport saturates at real human exposures depends on which parameter family you believe. There are six independent in vitro K_m values for PFOA against human transporters - 47 to 310 μM , from two laboratories, agreeing within a factor of 7 - against a PBPK transport-affinity constant of **0.133 μM** , which is 354 to 2,336× lower than anything ever measured in a cell. The in vitro values put human serum far below half-saturation; the fitted value puts three of four human populations at or above it, which would make clearance dose-dependent in contaminated communities and mean a single clearance factor cannot transfer between exposure settings. The weight of evidence has moved onto the in vitro side - the fitted values are fitted to plasma curves rather than measured, the mouse row has standard errors exceeding its estimates, and the dose-response evidence agrees with the in vitro answer - but nobody has measured a human value, and that single number would close it.

2. Unbound fraction by one method across species

Every value of R is proportional to f_u , and the mouse and human numbers currently come from different methods at different ligand:protein ratios. Measuring them one way decides whether the species difference is binding or transport - a 33× swing in the answer. Plasma, a dialysis or ultrafiltration rig, LC-MS/MS; no animals required. This is the cheapest decisive experiment the work identifies.

3. Whether the biliary 0.97 replicates beyond n = 4

It is now load-bearing for the human limb of finding 2, and it rests on four surgical patients in a single 2007 study.

One formerly-open conflict is now closed

The roughly 100× disagreement over PFOA's plasma free fraction was not a contradiction between two measurements. Han 2003's ">90% bound" is a *calculation* from K_d and albumin concentration, and it is a floor that Fischer's measured 0.00061 satisfies. The underlying K_d gap is a ligand:protein ratio artefact - Han titrated at 1.7:1 to 60:1, Fischer at 0.004:1 or below, and human serum sits at 10^{-5} to 10^{-3} :1. The defect was in the inheritance: PBPK models read ">90%" as "about 90%" and used a free fraction roughly **164× too high**.

What exists, and how to check it

db/combined/	1,438 rows in four schemas; every row carries its provenance and primary source. Also one Excel workbook.
db/primary_2026/	24 per-paper extractions, read from the papers directly
db/qsar/	per-compound structure descriptors paired with the renal handling ratio of finding 7
scripts/	34 runnable scripts - every figure and table above regenerates
figures/	16 figures
report/REPORT.md	the full review, about 2,002 lines
WANTED.md	the papers still unobtainable, and what each would change

Nothing in this document is quoted from memory. Where a value came from another paper's citation rather than the original, the database row says so. Where a value was computed in this work rather than read from a paper, the row says that too. Where this work's own method is known to be biased, the affected rows carry a flag and the report says to read section 8 item 9 before using them.

This PDF is generated by scripts/make_summary_pdf.py, which reads the reabsorption axis, the structure-activity table and every count above from the database at build time - so the document cannot drift from the data it describes.